

Jerry Liu

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Research Focus I'm interested in building efficient memory architectures for language models. My recent work studies how in-weights knowledge storage scales with model parameters and how in-context memory can be represented efficiently and reliably in recurrent-state architectures. My broader research background spans applied mathematics, numerical methods, and AI for science.

EDUCATION

Stanford University — Ph.D. Candidate, Computational & Mathematical Engineering Sep 2022 — Present
Advisor: Chris Ré @ Hazy Research | DOE Computational Science Graduate Fellow Expected graduation: Late 2026 / Early 2027
Duke University — B.S. Mathematics & Computer Science, Minor in Music Aug 2018 — Dec 2021

RESEARCH & INDUSTRY EXPERIENCE

Stanford University, Hazy Research — PhD Researcher Sep 2022 — Present
Advisor: Chris Ré

- Developed an information-theoretically optimal fact-storing MLP construction for Transformers, using **10–104× fewer parameters** than prior work at matched fact counts (COLM 2026, co-first author). Demonstrated proof-of-concept **modular fact editing** by swapping a Transformer's MLP without retraining.
- Leading ongoing work on recurrent architectures with modular recall components designed to improve in-context retrieval. Current results **match attention baselines in perplexity** at ~360M parameters by **scaling recurrent state size**.
- Developed sequence mixer architectures and a high-precision training recipe for in-context gradient descent on least-squares problems, achieving **100,000× lower MSE** and a **10,000× smaller out-of-distribution generalization gap** than standard Transformers (ICLR 2025).
- Designed BWLer, a new architecture for physics-informed neural networks (PINNs) that decouples PDE solution parameterization from derivative computation and **reduces RMSE by up to 1,800×** (TASC @ COLT 2025 Best Paper).

Lawrence Berkeley National Lab, Machine Learning & Analytics — Research Intern Jun 2023 — Sep 2023
Advisors: N. Benjamin Erichson, Michael W. Mahoney

- Explored Transformer in-context learning for solving differential equations. Developed a theoretical reduction of in-context operator learning to least squares; built a pseudospectral-solver-inspired implementation and evaluated robustness to distribution shifts (DLDE III @ NeurIPS 2023).

Duolingo, Learning AI Lab — Software Engineer Jan 2022 — Sep 2022

- Trained and maintained a multilingual user-knowledge LLM serving **9 million daily active users**. Productionized data-cleaning and retraining pipelines for evolving features and data, and built performance diagnostics.

Lawrence Livermore National Lab — Research Intern Jun 2021 — Dec 2021
Advisors: Jin Yao, Kenneth Weiss

- Developed and implemented high-order numerical algorithms for geometric interface reconstruction and volume advection in incompressible, multi-material fluid simulations.

SELECTED PUBLICATIONS

- Roberto Garcia*, **Jerry Liu***, Ronny Junkins*, Sabri Eyuboglu, Atri Rudra, Chris Ré. *MLPs are Hebbians: Constructing Efficient Fact-Storing MLPs for Transformers*. COLM 2026. [[paper](#) | [code](#) | [blog](#)]
- Jerry Liu**, Jessica Grogan, Owen Dugan, Ashish Rao, Simran Arora, Atri Rudra, Chris Ré. *Towards Learning High-Precision Least Squares Algorithms with Sequence Models*. ICLR 2025. [[paper](#) | [code](#)]
- Jerry Liu**, Yasa Baig, Denise Hui Jean Lee, Rajat Vadiraj Dwaraknath, Atri Rudra, Chris Ré. *BWLer: Barycentric Weight Layer Elucidates a Precision-Conditioning Tradeoff for PINNs*. TASC @ COLT 2025 (**Best Paper Award**) [[paper](#) | [code](#) | [blog](#)]
- Catherine Deng*, Junmiao Hu*, Milan Rohatgi, **Jerry Liu**, Chris Ré. *Constructing Machine-Precision Neural Networks with Quasi-Interpolants*. AI&PDE @ ICLR 2026 **Oral (8/113 accepted papers)**. [[paper](#)]
- Jerry Liu**, N. Benjamin Erichson, Kush Bhatia, Michael W. Mahoney, Chris Ré. *Does In-Context Operator Learning Generalize to Domain-Shifted Settings?* DLDE III, NeurIPS 2023. [[paper](#)]

* Equal contribution.

SKILLS

ML & Computing Python, C++; PyTorch, JAX, Hugging Face Transformers, Triton; NumPy, SciPy; Slurm
Mathematics Numerical linear algebra, optimization, numerical PDEs, approximation theory, probability & statistics